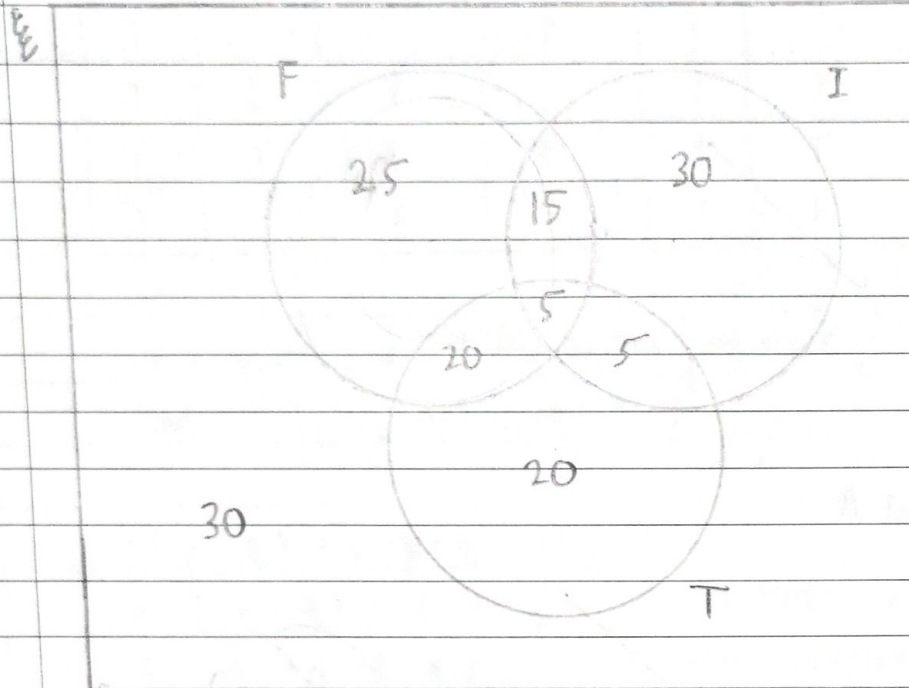


Question 1

- a) i) F = FC students that obtain Facebook
 I = FC students that obtain Instagram
 T = FC students that obtain Twitter



Total FC student = 150
 Total Facebook users = 65
 Total Instagram users = 55
 Total Twitter users = 50

ii) $(F \cup I \cup T)' = 150 - (25 + 15 + 5 + 5 + 20 + 30 + 20)$
 $= 30$

$\therefore$ There are 30 students that do not have an account for three social media.

iii) Student that exactly have two social media
 $= 15 + 5 + 20$
 $= 40$

$\therefore$ There are 40 students that have exactly two social media. (8)

iv) Students that have social media other than Facebook
 $= 20 + 5 + 30$
 $= 55$

$\therefore$ There are 55 students that have social media other than Facebook.

$$b) A = \{3, 5, 7, 9\}$$

$$B = \{2, 3, 5, 7\}$$

$$C = \{3, 6, 9\}$$

$$i) |A| = 4$$

$$|B| = 4$$

$$|C| = 3$$

$$ii) |A| = 4$$

Number of proper subset A

$$= 2^n - 1$$

$$= 2^{(4)} - 1$$

$$= 16 - 1$$

$$= 15$$

$$iii) C \times B = \{(3, 2), (3, 3), (3, 5), (3, 7), (6, 2), (6, 3), (6, 5), (6, 7), (9, 2), (9, 3), (9, 5), (9, 7)\}$$

Question 2

2)

(a) Truth Table

P	q	$p \vee q$	$\neg(p \vee q)$	$\neg p$	$(\neg p \wedge q)$	$\neg(p \vee q) \vee (\neg p \wedge q)$
T	T	T	F	F	F	F
T	F	T	F	F	F	F
F	T	T	F	T	T	T
F	F	F	T	T	F	T

Since column 5 and 7 showing the same result,
 thus, $\neg(p \vee q) \vee (\neg p \wedge q) \equiv \neg p$

$$\begin{aligned}
 & \neg(p \vee q) \vee (\neg p \wedge q) \\
 & \equiv (\neg p \wedge \neg q) \vee (\neg p \wedge q) \\
 & \equiv \neg p \wedge (\neg q \vee q) \\
 & \equiv \neg p \wedge T \\
 & \equiv \neg p \quad (\text{proven})
 \end{aligned}$$

De Morgan's Laws

Distributive Laws

~~Absorption Laws~~ Complement Laws

(b) (i) $(r \wedge q) \rightarrow p$

(ii) $\neg(r \vee q) \rightarrow \neg p$
 $\equiv (\neg r \wedge \neg q) \rightarrow \neg p$

(iii) $\neg p \rightarrow \neg(r \vee q)$
 $\equiv \neg p \rightarrow (\neg r \wedge \neg q)$

(c) $\neg[\forall x (x^2 + 2x - 3 = 0)]$
 $\equiv \exists x (x^2 + 2x - 3 \neq 0)$

If $x=2$,

$(2)^2 + 2(2) - 3 = 5$

$(\neq 0)$

If $x=3$,

$(3)^2 + 2(3) - 3 = 12$

$(\neq 0)$

$\therefore x^2 + 2x - 3 \neq 0$ is true for some values of x .
 the statement is true.

(d) let $P(x)$ = students who can speak Russian

$Q(x)$ = students who know C++

with the domain of discourse is consist of all students at my school.

(i) $\exists x (P(x) \wedge \neg Q(x))$

(ii) $\forall x (P(x) \vee Q(x))$

(iii) $\forall x (\neg P(x) \wedge \neg Q(x))$

Question 3

(a) For all integer, if $a^2 - 3b$ is even then a is even and b is even.

$$p(x) = a^2 - 3b \text{ is even}$$

$$q(x) = a \text{ is even and } b \text{ is even}$$

$$p(x) \rightarrow q(x) \equiv \sim q(x) \rightarrow \sim p(x)$$

indirect proof: $\sim q(x) \rightarrow \sim p(x)$

Assume $\sim q(x)$ is true,

a is odd or b is odd,

Case 1: let a is odd, b is even,

$$a = 2m + 1$$

$$b = 2n$$

$$a^2 - 3b = (2m + 1)^2 - 3(2n)$$

$$= 4m^2 + 4m + 1 - 6n$$

$$= 2(2m^2 + 2m - 3n) + 1$$

$$\text{where } t = 2m^2 + 2m - 3n$$

$$= 2t + 1 \quad (\text{integer})$$

$2t + 1$ is odd integer.

$\therefore$ If $\sim q(x)$ is true, then $\sim p(x)$ is true. Thus,

$\sim q(x) \rightarrow \sim p(x)$ is true.

Case 2: Assume $\sim q(x)$ is true.

If a is even, b is odd.

$$a = 2x$$

$$b = 2y + 1$$

$$a^2 - 3b = (2x)^2 - 3(2y + 1)$$

$$= 4x^2 - 6y - 3$$

$$= 4x^2 - 6y - 4 + 1$$

$$= 2(2x^2 - 3y - 2) + 1$$

$$\text{where } l = 2x^2 - 3y - 2,$$

$$= 2l + 1 \quad (\text{integer})$$

$2l + 1$ is odd integer

$\therefore$ If $\sim q(x)$ is true then $\sim p(x)$ is true. Thus,

$\sim q(x) \rightarrow \sim p(x)$ is true.

Assume $\sim q(x)$ is false.

Case 2:

let a is even, b is even

$$a = 2m$$

$$b = 2n$$

$$a^2 - 3b = (2m)^2 - 3(2n)$$

$$= 4m^2 - 6n$$

$$= 2(2m^2 - 3n)$$

$$\text{where } k = 2m^2 - 3n$$

(integer)

$$= 2k$$

$\therefore 2k$ is even integer. $\sim q(x)$ is false, $\sim p(x)$ is false, $\sim q(x) \rightarrow \sim p(x)$ is true.

Case 3: Assume $\sim q(x)$ is true.

If a is odd, b is odd.

$$a = 2j + 1$$

$$b = 2k + 1$$

$$a^2 - 3b = (2j + 1)^2 - 3(2k + 1)$$

$$= 4j^2 + 4j + 1 - 6k - 3$$

$$= 4j^2 + 4j - 6k - 2$$

$$= 2(2j^2 + 2j - 3k - 1)$$

$$\text{where } f = 2j^2 + 2j - 3k - 1$$

$$= 2f \quad (\text{even integer})$$

(false)

If $\sim q(x)$ is true, then $\sim p(x)$ is false.

Thus $\sim q(x) \rightarrow \sim p(x)$ is false.

Ans: The statement is false.